

GRAVITYSORT · DESIGN DECISIONS REGISTER

GravitySort design decisions register

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DOCUMENT CONTROL

Document ID	GVS-DEC-001	Version	0.6
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/gravitysort		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-02	Amish Chadha	Decisions of 2026-10-02 carried into the model, BOM, calculations and build plan; one open decision added (tank on the centre line and the tipping criterion, made to carry out decision 1); Value engineering updated to USD 580
0.5	2026-10-02	Amish Chadha	Amish approved the recommendations for open decisions 1 to 5; moved to decisions made
0.4	2026-10-01	Amish Chadha	Amish accepted the change made while adding the bump stop (slotted pitman pin, leaning legs; GVS-DDR-003 Table 2); added to decisions made; no open item changed
0.3	2026-10-01	Amish Chadha	Table bump stop now in the model and build plan (follow-up note removed); buffer added to the items to confirm; Value engineering updated to USD 569
0.2	2026-10-01	Amish Chadha	Amish accepted the recommendations of open items 1 to 4, 9, 10 and 12 (GVS-DDR-003 accepted; R11

restated); moved to decisions made;
open items renumbered 1 to 5

0.1

2026-10-01

Amish Chadha

Register opened with the build plan;
open decisions from the review notes,
the decision records and the design
for construction

GravitySort design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md` , GVS-BLD-001) describes the design as it stands and does not list open decisions.

■ Open decisions

#	To decide	Options	Recommendation	What it affects in the build	Source
1	Where the 1.7 m tank post stands, and what "a full tank cannot tip it" means. To carry out the decision of 2026-10-02 the post was moved from 180 mm behind the centre line onto it, and the check uses a 10 degree slope in any direction with a full tank. On the centre line the machine tips at 13.5 degrees (193 N side push at the tank); left 180 mm behind, it would tip at 9.5 degrees (133 N)	(a) keep the post on the centre line and the 10 degree criterion, as made; (b) put it back 180 mm behind and add wider feet or outriggers at the back; (c) a stricter criterion (for example 15 degrees), which needs wider feet as well	(a): no extra parts, better than the old 1.25 m layout (11.6 degrees)	Frame (tank post, braces, cradle), hose route, build plan section 3.1 and step 13	GVS-CAL-001 v0.8 section 11; review note 2026-10-02

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The two flanged bearing units (UCF205 class): 95 mm square flange, bolts 70 mm apart, 25 mm bore with set screws	They set the holes in the two bearing plates	GVS-DDR-003, P2
2	The shaft flange hub: 25 mm bore, 80 mm flange, four holes on a 60 mm circle, room for a 6 mm cross pin	It joins the bowl and jacket to the spindle	GVS-DDR-003, P4
3	The rotary union: 1/2 in BSP on the shaft side, side water port, seal drag near 0.10 N m	Pedal power (R6) assumes this drag; the nipple welded into the spindle must match its thread	GVS-CAL-001 sections 7 and 10
4	The right-angle 1:1 gearbox: through input shaft about 33 mm above its feet, feet about 100 x 80 mm, output shaft long enough for the drive pulley 85 mm below the feet	It sets the jackshaft height and the member spacing	GVS-DDR-003, P9; GVS-PRC-001 open questions
5	The four 20 mm pillow blocks (UCP204 class, centre height 33.3 mm, bolts 95 mm apart)	Two set the jackshaft height on the gearbox members; two carry the table head shaft	GVS-DDR-003, P8 and P14
6	The two rubber pipe grommets for 63 mm and 75 mm pipe that seal in a 6 mm curved wall	They seal the standpipe and tailings pipe without welding	GVS-DDR-003, P6
7	The 430 mm HDPE drum: flat floor, 6 mm wall, food-grade or clean	It is the splash tub	GVS-DDR-003, P6
8	The used bicycle parts: a bottom bracket shell that can be cut from a scrap frame and welded, crank axle length, a 27.2 mm seat post	They set the outrigger's pedal post and seat tube	GVS-DDR-003, P8
9	The disc caliper's mount (post mount or the older standard) and the disc's 6-bolt pattern	They set the caliper bracket holes and the brake flange	GVS-DDR-003, P15
10	MotionCore: the reference hub motor's no-load speed (assumed 250 rpm), a sprocket on its disc mount, axle length for dropouts 76 mm apart inside, a speed limit that maps to 900 rpm at the spindle, a brake input for the lid switch	They set the 1.2:1 step-up, the cradle and the motor speed cap	GVS-DDR-002, item 14; review note 2026-09-25
11	The bump stop's rubber buffer: 40 mm across, 30 mm long, about 55 Shore A, M8 female thread, rated 800 N or more in compression	The stop's force, squeeze and setting range are sized for this buffer (GVS-CAL-001 section 9)	GVS-DDR-003, A2

■ Value engineering

Value-engineering target: USD 455 (a hypothetical control target, not a limit; Amish, 2026-10-01). Estimated cost of the constructable design: USD 580 (USD 125 over the target), excluding the MotionCore kit, reference motor and battery. Main cost drivers and savings worth trying:

- **Main cost drivers.** The jackshaft, gearbox, chains and pulleys (\$60); the jacket, hub and rotary union (\$55); the table stand and head (\$52); the frame (\$47); the bowl (\$46); the water supply (\$41). Making the design buildable added \$103, mostly the table head and tensioner, more frame tube and plate, and the tub fittings

(GVS-DDR-003); the table bump stop adds \$11 more (bracket and striker \$6, buffer and fixings \$5); the decisions of 2026-10-02 add \$11 (the braced 1.7 m tank post \$4, the flush container with its hasp \$7).

- **A quarter-turn belt instead of the bevel gearbox**, about \$25 less (GVS-DDR-001, item 1). Worth a test, since a used gearbox is also the hardest part to find.
- **Used bearing units and pillow blocks** from scrap farm or factory machinery, about \$10 to \$15 less across the six bearings.
- **Run the table head shaft in two plain bronze bushes** in the head plate instead of two pillow blocks, about \$8 less; the shaft turns slowly (about 300 rpm).
- **Take the table wash water from the main hose** with a tee at the rotameter instead of a separate branch and valve, about \$4 less.
- **Buy steel as offcuts** and use one plate thickness (5 mm) throughout, a few dollars and a little mass.
- Not recommended: the non-fluidized bowl (about \$45 less) or building the table later (about \$100 deferred), because both change what the machine does.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items: two stages, fluidized bowl with rotary union, cast PU liner, pedal baseline with the MotionCore option and one time-shared drive, battery left to the user, borax smelting outside the machine, partner screening criterion	Amish: "i accept all your recommendations, go with them across all repos."	GVS-DDR-001, GVS-DDR-002
2026-09-25	Budget from \$350 to \$450; R12 restated for the pedal drive; frame in 25 x 25 x 1.5 mm tube; 3/4 in fluidization hose and graded holes; disc brake with parking latch and lid interlock pin, speed display, 1.2:1 motor step-up, MotionCore lid switch	Amish, same instruction	GVS-DDR-002, items 1, 9, 10, 11a and 14
2026-09-26	Budget set to \$455 to cover the priced BOM	Amish: "i approve all the budget items."	GVS-DDR-002 v0.2
2026-09-26	GravitySort chosen for the first batch of product renders	Amish	docs/ REVIEW.md , session of 2026-09-26
2026-09-30	Make the design physically buildable while drawing the build plan; the changes are recorded in GVS-DDR-003 and were accepted on 2026-10-01 (below)	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	GVS-DDR-003
2026-09-30	Open decisions are kept out of the build plan, in this register	Amish: "don't log outstanding decisions in this build plan - that is not the place for it. that should be in a separate design document logged and named as such"	.kit/ STANDARDS.md section 18
2026-10-01	The budget is a value-engineering target, not a limit; cost is reported against it	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	.kit/ STANDARDS.md section 18
2026-10-01	Design for construction accepted: the changes P1 to P15 and their knock-on changes, as made	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	GVS-DDR-003, Tables 1 and 2
2026-10-01	R11 restated: total 100 kg or less, every load 30 kg or less (was 80 kg in all); the 95.7 kg design meets it on paper. Follow-up: try the savings of about 16 kg (option b) at TRL 4	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	GVS-DDR-003, A1; GVS-REQ-001 v0.7
2026-10-01	Asymmetric table stroke: an adjustable rubber bump stop at the return end of the stroke for the prototype. Follow-up: a toggle head (option b) only if the first test shows the stop is not enough	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	GVS-DDR-003, A2

Date	Decision	Decided by	Record
2026-10-01	Pedal position: keep the bicycle-style seat, about 910 to 960 mm up, seat post adjustable. Follow-up: check with two riders at TRL 4	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	GVS-DDR-003, A3
2026-10-01	Containment of a liner fragment: check by calculation before any spin test; add a steel band round the tub at the bowl lip height only if the check fails	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	GVS-CAL-001 section 10; review note 2026-09-25
2026-10-01	Lid: keep the plain 10 mm HDPE lid, no sight window	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	Review note 2026-09-26, item 4
2026-10-01	Renderers: the cranks turned 100 degrees from top dead centre is a render pose only; the model keeps its crank angle	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	Review note 2026-09-26, item 1
2026-10-01	Table bump stop, change made to let the deck strike the stop: the pitman pin works in a 12 x 20 mm slot in the deck cheeks (8 mm of free play), and the plywood legs are set leaning about 5 mm so they press the deck on the buffer with about 200 N. Accepted as made; no design change	Amish: "I approve of your recommendations for PicoFlow and GravitySort"	GVS-DDR-003, Table 2 (v0.4)
2026-10-02	Head for the fluidization supply: raise the header tank post to 1.7 m (option a), braced so a full 60 L tank cannot tip it; no pump on the fluidization line. Carried into the model, drawings, build plan, BOM and GVS-CAL-001 v0.8 on 2026-10-02 (8.6 kPa at the union at mid-tank)	Amish: "i approve your recommendations for all 555 open decisions."	GVS-DDR-001, item 11b
2026-10-02	Water at pedal-only sites: gravity supply from upstream is the site rule for the first field trial; where that is impossible, a bought treadle or hand pump worked in turns by the crew, or a small 12 V pump at sites with the MotionCore battery	Amish: "i approve your recommendations for all 555 open decisions."	GVS-DDR-001, item 12
2026-10-02	R2 restated as 200 kg/h of feed time, about 1.55 t in an eight-hour shift with three flush stops; the design feed stays at 200 kg/h	Amish: "i approve your recommendations for all 555 open decisions."	GVS-DDR-001, item 13
2026-10-02	First co-design partner and country: Colombia, where the mercury ban is in force, with a miners' cooperative introduced through the Alliance for Responsible Mining in Medellin as the first candidate to approach	Amish: "i approve your recommendations for all 555 open decisions."	GVS-DDR-001, item 8b
2026-10-02	Concentrate security: padlock hasps on both the concentrate box and the flush container	Amish: "i approve your recommendations for all 555 open decisions."	GVS-PRC-001, open questions

Date	Decision	Decided by	Record
	as the baseline, and a two-person rule for opening them proposed to the partner; both to be confirmed with the partner. Flush container with hasp added to the model and BOM (item 23) on 2026-10-02		